Exercise Sheet 24

The notation is that of Lecture 24. The adapted basis is e_i, f_i, with u_0 in odd dimension, and $S = \Lambda W$. The operators μ_i, ι_i and θ are the creation, annihilation and parity operators. We use

$$M_{u,v}x = 2q(v,x)u - 2q(u,x)v, \qquad N_i = \mu_i\iota_i, \qquad \rho(H_i) = N_i - \frac{1}{2}.$$

Exercises 24.1–24.3 are routine calculations. Exercises 24.4–24.6 reconstruct and extend proofs.

Exercise 24.1 (Creation and annihilation matrices). Let $n = 2$ and use the ordered basis $(b_0, b_1, b_2, b_3) = (1, e_1, e_2, e_1 \wedge e_2)$ of S. For $A \in \mathfrak{so}(V)$, write $D(A)$ for the matrix of $\rho(A)$. For an auxiliary operator T on S, write $D(T)$ for its matrix in the same basis.

(i) Write the matrices $D(\mu_1)$, $D(\mu_2)$, $D(\iota_1)$, $D(\iota_2)$ and $D(\theta)$.

(ii) Verify $\mu_2\iota_2 + \iota_2\mu_2 = \mathrm{id}$ and $\mu_1\iota_2 + \iota_2\mu_1 = 0$ using the matrices from (i).

(iii) Compute $D(H_1)$ and $D(H_2)$.

Exercise 24.2 (The spin module of $\mathfrak{so}_5$). Let $N = 5$, so $n = 2$ and V has basis e_1, e_2, u_0, f_1, f_2. Use the basis of Exercise 24.1 for S.

(i) List the weights of $b_0, \dots, b_3$.

(ii) The simple roots of B_2 are $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2$. Compute the matrices of the operators $\rho(X_1)$, $\rho(X_2)$, $\rho(Y_1)$ and $\rho(Y_2)$, where $X_1 = M_{e_1,f_2}$, $X_2 = M_{e_2,u_0}$, $Y_1 = M_{e_2,f_1}$ and $Y_2 = M_{f_2,u_0}$.

(iii) Show directly that S is irreducible with highest weight $\omega_2 = \frac{1}{2}(\varepsilon_1 + \varepsilon_2)$.

(iv) Check $\dim L(\omega_2) = 4$ with the Weyl dimension formula.

Exercise 24.3 (Half-spin modules of $\mathfrak{so}_4$). Let $N = 4$. Then $S^+ = \mathrm{span}\{1, e_{12}\}$ and $S^- = \mathrm{span}\{e_1, e_2\}$.

(i) Show that $E = \rho(M_{e_1,e_2})$, $H = \rho(H_1 + H_2)$ and $F = -\rho(M_{f_1,f_2})$ act on S^+ as an $\mathfrak{sl}_2$-triple, and act by zero on S^-.

(ii) Show that $E' = \rho(M_{e_1,f_2})$, $H' = \rho(H_1 - H_2)$ and $F' = \rho(M_{e_2,f_1})$ act on S^- as an $\mathfrak{sl}_2$-triple, and act by zero on S^+.

(iii) Deduce $\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$, with S^+ and S^- the defining modules of the two factors.

Exercise 24.4 (Sign and normalisation conventions).

(i) Another common identification uses $M'_{u,v}x = q(v,x)u - q(u,x)v$. Find the constant c with $[c(uv - vu), x] = M'_{u,v}x$, and compare the resulting map $\mathfrak{so}(V) \to \mathrm{Cl}(V)$ with σ.

(ii) Let $\tau = -\sigma$. Show that $[\tau(A), x] = -Ax$ and that τ is not a Lie algebra homomorphism.

Exercise 24.5 (Spinors of $\mathfrak{so}_8$ and the weight lattice). Let $N = 8$, so $n = 4$.

(i) List the weights of S^+ and S^- and confirm both dimensions.

(ii) With roots $\pm\varepsilon_i \pm \varepsilon_j$ and $\alpha^\vee = \alpha$, show that $P = \mathbb{Z}^4 \cup \big(\mathbb{Z}^4 + \frac{1}{2}(1,1,1,1)\big)$ and $Q = \{a \in \mathbb{Z}^4 : \sum a_i \text{ even}\}$ in ε-coordinates.

(iii) Show that $\omega_1 = \varepsilon_1$, ω_3 and ω_4 lie in the three distinct nonzero cosets of Q in P.

Exercise 24.6 (All simple orthogonal modules from fundamental ones). Let $\mathfrak{g}$ be semisimple, and let $\lambda, \mu \in P_+$.

(i) Prove that $L(\lambda + \mu)$ is a direct summand of $L(\lambda) \otimes L(\mu)$.

(ii) Deduce that every finite-dimensional simple $\mathfrak{so}_{2n+1}$-module is a summand of a tensor product of copies of $\Lambda^k V$ $(1 \leq k \leq n-1)$ and S. State the analogous result for $\mathfrak{so}_{2n}$.